

A PROJECT REPORT ON

MULTITENANT LEAVE SYSTEM

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FOR THE AWARD OF THE DEGREE OF

BACHELOR OF ENGINEERING (COMPUTER ENGINEERING)

SUBMITTED BY

AKSHAT DHANUKA
AAYUSH BHASKARWAR
ZIYAAD PARKAR

Exam No : B400050317
Exam No : B400050338
Exam No : B400050640



DEPARTMENT OF COMPUTER ENGINEERING

PUNE INSTITUTE OF COMPUTER TECHNOLOGY
DHANKAWADI, PUNE 411043

SAVITRIBAI PHULE PUNE UNIVERSITY
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CERTIFICATE

This is to certify that the project report entitled

“MULTITENANT LEAVE SYSTEM”

Submitted by

AKSHAT DHANUKA

AAYUSH BHASKARWAR

ZIYAAD PARKAR

Exam No : B400050317

Exam No : B400050338

Exam No : B400050640

are bonafide students of this institute and the work has been carried out by them under the supervision of **Prof. S.P. Shintre** and it is approved for the partial fulfillment of the requirement of Savitribai Phule Pune University, for the award of the degree of Bachelor of Engineering (Computer Engineering).

(Prof. S.P. Shintre)

Guide,

Department of Computer Engineering

(Dr. G. V. Kale)

Head,

Department of Computer Engineering

(Dr. S. T. Gandhe)

Principal,

Pune Institute of Computer Technology

Place : Pune

Date :

Certificate of Sponsorship



Fifth Floor, Parakh Capital,
Hadapsar Pune - 411028
+91 9529556802 / 7030089090
info@techbsoftwares.com
www.techbsoftwares.com

TO WHOM IT MAY CONCERN

We are pleased to confirm that TechB Softwares LLP will sponsor the final-year project titled Multi-Tenant Leave Software done by students:

1. Akshat Dhanuka,
2. Aayush Bhaskarwar
3. Ziyaad Parkar.

Our sponsorship will only include technical guidance from our team, led by Mehul Agrawal who will also act as the students mentor throughout the course of the project.

Yours sincerely,

For and on behalf of TechB Softwares LLP.

A handwritten signature in blue ink, appearing to read 'Mehul', with a stylized flourish extending from the end.

Mehul Agrawal

Founder & CEO.

TechB Softwares LLP.

ACKNOWLEDGEMENT

It gives us great pleasure in presenting the project report on “**MULTITENANT LEAVE SYSTEM**”.

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**AKSHAT DHANUKA
AAYUSH BHASKARWAR
ZIYAAD PARKAR**

SCTR's PICT, Department of Computer Engineering 2024-25

ABSTRACT

In today's complex enterprise environments, effective leave management is critical for maintaining productivity while ensuring employee well-being. This project presents a Multi-Tenant Leave Management System, designed to serve diverse organizational units with configurable leave policies, automated decision-making, and intelligent recommendations. The system accommodates multiple departments or clients (tenants) under a unified platform, enabling tailored rule sets, secure access, and efficient workflows.

To optimize resource planning and leave utilization, the system integrates advanced analytics through custom algorithms. The ALO-60 (Adaptive Leave Optimization) algorithm dynamically adjusts sick leave quotas based on historical usage, reducing over-allocation and promoting fairness. The Utilization KPI framework analyzes leave trends per activity code and department, identifying underused rules for consolidation and flagging overused ones for policy expansion. These insights ensure compliance while simplifying policy management.

Further enhancing the platform's intelligence, the PLRA-I (Proactive Leave Recommendation Algorithm) evaluates historical leave behavior, peak workloads, and team availability to suggest ideal leave periods for employees. Integration with AI models like Gemini enables real-time analysis and personalized insights. Together, these components transform traditional leave systems into a data-driven, employee-centric solution that supports strategic planning, operational continuity, and improved work-life balance.

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LIST OF ABBREVIATIONS

- HRMS – Human Resource Management System
- KPI – Key Performance Indicator
- ALO-60 – Adaptive Leave Optimization – 60% Threshold
- PLRA-I – Proactive Leave Recommendation Algorithm – Insight
- UI – User Interface
- UX – User Experience
- DBMS – Database Management System
- API – Application Programming Interface
- SQL – Structured Query Language
- FTP – File Transfer Protocol
- JWT – JSON Web Token
- OTP – One-Time Password

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CHAPTER 1

INTRODUCTION

1.1 Overview :

This project is a **Multi-Tenant Leave Management System** designed for companies with multiple departments or clients. It allows each tenant to manage their leave rules, workflows, and users independently. The system includes smart algorithms to track leave usage, recommend changes in policies, and suggest the best time for employees to take leave. It helps both HR teams and employees by reducing manual work, improving accuracy, and supporting smarter planning using data and AI.

1.2 Motivation :

In many companies, leave systems are either too basic or too complex, often leading to misuse, unbalanced workloads, or missed leaves. As work environments become more dynamic, there is a growing need for a system that can adapt to different teams and make smart suggestions based on real data. This project was motivated by the idea of combining traditional leave tracking with AI-driven insights, improving employee experience while also helping organizations plan better and stay productive.

1.3 Problem Definition & Objectives :

Traditional leave management tools often fail to adapt to multiple departments, lack smart decision-making, and don't provide insights based on actual usage or productivity. There is no built-in logic to suggest when employees should take leave or how to optimize leave policies. Our project aims to solve this by creating a smart, flexible, and multi-tenant system that not only tracks leaves but also analyzes data to improve policies and support decision-making for both employees and managers.

Objectives:

- Enable Multi-Tenant Functionality: Allow different departments or client organizations to manage their own leave policies, workflows, and users independently within the same platform.
- Analyze Leave Utilization: Use historical data to calculate KPIs and recommend policy changes for better efficiency and compliance.
- Integrate Smart Algorithms: Implement ALO-60 and PLRA-I to optimize sick leave allocation and proactively recommend leave periods based on workload and demand. Lastly, Simplify the leave application and approval process for employees and managers through a responsive, data-driven interface.

1.4 Project Scope & Limitations :

The system relies on accurate attendance and productivity data, which may not always be consistently available. AI-driven recommendations might not fully capture human factors like burnout or personal emergencies. Additionally, real-time performance can vary depending on data volume and the availability of APIs like Gemini or Twitter.

1.5 Methodologies of Problem Solving :

To solve the identified problem, we implemented a modular pipeline involving three core components:

- Data-Driven Analysis: Historical leave and productivity data are examined to detect usage patterns and areas of inefficiency.
- AI and NLP Integration: Algorithms such as ALO-60 and PLRA-I, supported by Gemini AI, are used to optimize leave planning and recommendations.
- KPI-Based Optimization: Utilization KPIs are calculated for each activity code to support decisions like increasing, merging, or retiring specific leave rules.

CHAPTER 2

LITERATURE SURVEY

1. Cloud-Based Employee Leave Management System [1]

Summary:

This paper[1] presents the design and implementation of a web-based Employee Leave Management System aimed at automating the leave application and approval process within organizations. The system enables employees to submit leave requests, view leave balances, and track the status of their applications. Administrators can manage leave policies, approve or reject requests, and generate reports. The study emphasizes the transition from manual, paper-based processes to a streamlined digital solution, highlighting improvements in efficiency, accuracy, and record-keeping. The system's architecture supports scalability and can be adapted to various organizational needs, making it a foundational reference for developing multi-tenant leave management applications.

2. Design and Implementation of a Web Based Leave Management System [2]

Summary:

This paper[2] presents a web-based leave management system aimed at improving administrative efficiency. Developed using PHP and MySQL, the system provides functionalities like user authentication, leave application, approval workflow, and report generation. It emphasizes reducing paperwork and decision delays. The authors also highlight the importance of user-friendly interfaces and real-time data access in enhancing productivity.

3. Review on Leave Management Systems [3]

Summary:

This paper[3] provides a comprehensive review of existing LMS solutions, comparing traditional manual systems with automated software. It identifies major challenges like record maintenance, approval tracking, and leave balance computation. The review advocates for web-based and mobile-

integrated systems that offer notification services and real-time updates, thereby improving organizational transparency and employee satisfaction.

4. Leave Management System [4]

Summary:

This paper[4] proposes a lightweight leave management system tool tailored for educational institutions. It integrates a simple GUI with database connectivity to manage student and faculty leave records. The system's key contributions include tracking leave history, automated notifications, and a streamlined approval process. Its design targets ease of use, particularly for non-technical users.

5. Management Planning and Employee Engagement: A Review of Literature [5]

Summary:

This paper[5] explores the relationship between management planning and employee engagement. It suggests that effective leave policies are a component of strategic HR planning, directly influencing staff motivation and retention. The paper underscores the importance of transparent leave practices as part of broader organizational planning frameworks.

6. Exploring Leave Management Practices and Relationship with Performance of Administrative Staff: Evidence from a Tertiary Institution in Ghana [6]

Summary:

This paper[6] investigates how leave practices affect staff performance in academic institutions. Using survey methods, it identifies a positive correlation between well-structured leave systems and administrative efficiency. The study recommends adopting automated systems to enhance fairness, consistency, and performance tracking.

7. Employee Leave Management System [7]

Summary:

This paper[7] focuses on the development and implementation of a computerized leave management system specifically for educational institutions. The study highlights the challenges faced with manual systems, such as loss of records and inefficiencies. The proposed system uses a structured database and a web-based interface to streamline leave applications, approvals, and report generation. The work emphasizes cost-effectiveness, data integrity, and improved management decision-making.

8. Mobile HRM for Online Leave Management System [8]

Summary:

This paper[8] proposes a mobile-centric leave management system integrated within Human Resource Management (HRM) solutions. The system is designed for Android devices, allowing employees to submit leave requests and managers to approve or reject them via mobile apps. It leverages GPS and real-time notifications for better interactivity and accessibility. The authors argue that mobile integration significantly enhances usability and encourages timely actions from both staff and administrators.

9. Predicting Employee Absence from Historical Absence Profiles with Machine Learning [9]

Summary:

This paper[9] explores how machine learning models can be used to predict employee absenteeism. It analyzes historical leave data and employs algorithms such as decision trees, random forests, and support vector machines to forecast patterns of absence. This predictive insight can enhance HR

planning and improve the strategic allocation of resources. It presents a novel integration of AI into workforce management systems.

10. Chatbot Based Helpdesk for Government Employee and Department [10]

Summary:

This paper[10] presents a chatbot-integrated helpdesk system, designed to assist government employees in accessing departmental services, including leave management. The chatbot utilizes natural language processing (NLP) to interact with users and respond to queries related to leave balance, policy clarification, and application status. Although not purely an LMS, the system exemplifies how AI-driven interfaces can enhance user interaction and self-service within administrative tools.

CHAPTER 3

SOFTWARE REQUIREMENTS SPECIFICATION

3.1 Assumptions & Dependencies :

- It is assumed that users will have consistent access to the internet for interacting with the leave management portal and for triggering AI-driven insights.
 - The project depends on access to historical employee data (leave, attendance, and productivity), which may vary in availability and format across tenants.
 - It is also assumed that required backend and AI libraries (e.g., Pandas, NumPy, Scikit-learn, Google Generative AI, Matplotlib) are compatible with the deployment environment.
 - The system assumes that tenants will maintain reasonably structured leave policies and workflows to enable optimization through the provided algorithms.
- .

3.2 Functional Requirements :

3.2.1 System Feature 1: Leave Management Core

- Handles multi-tenant leave request and approval workflows.
- Supports CRUD operations on leave policies and tracks employee leave records.
- Validates requests based on team availability and predefined leave rules.

3.2.2 System Feature 2: Productivity and KPI Analysis Module

- Analyzes leave impact on employee productivity using pre- and post-leave trends.
- Calculates department-wise Utilization KPIs based on leave consumption patterns.
- Visualizes data insights using plots and summaries for admin review.

3.2.3 System Feature 3: AI recommendation Engine

- Implements ALO-60 and PLRA-I algorithms to optimize leave allocation and planning.
- Uses historical data and workload patterns to suggest personalized leave recommendations.
- Integrates Google Gemini API for trend insights, employee classification, and scenario prediction.

3.3 External Interface Requirements :

3.3.1 User Interfaces :

- Admin and Employee dashboards with role-based access.
- Graphs, tables, and summaries to view productivity, KPI stats, and AI suggestions.
- Options for superusers to download reports or adjust policies dynamically.

3.3.2 Hardware Interfaces :

- No specialized hardware required - compatible with any standard web-connected electronic device.

3.3.3 Software Interfaces :

- Google Generative AI (Gemini) API for insights and recommendations.
- Django backend with PostgreSQL or MySQL for data storage.
- Front-end technologies such as React.js for dynamic user interaction.

3.3.4 Communication Interfaces :

- Secure HTTPS protocols for all client-server communication.
- Authentication mechanisms (e.g., JWT or session-based) for secure login and access control.

3.4 Non-Functional Requirements :

3.4.1 Performance Requirements :

- The system should support real-time response for leave requests, analytics, and AI suggestions within 5–10 seconds.
- KPI analysis should complete processing for at least 500 employees within 1 minute.
- Recommendation algorithms should achieve at least 85% accuracy based on historical test data and productivity correlation.

3.4.2 Safety Requirements :

- The system must not store or expose sensitive employee details such as personal identifiers or salary data.
- All AI modules and KPI calculations must operate on anonymized or departmental data to ensure compliance.
- Data queries and resource usage must be regulated to prevent system overload or performance bottlenecks.

3.4.3 Security Requirements :

- Sensitive data like authentication tokens and admin credentials must be securely encrypted and stored using environment variables.
- The system must implement role-based access control to restrict data visibility and editing permissions.

- Secure communication protocols (e.g., HTTPS, SSL) must be used for all client-server interactions.

3.4.4 Software Quality Attributes :

- **Modularity:** Leave management, KPI analysis, and AI recommendation modules are independently developed for easy updates.
- **Scalability:** The system supports adding new departments, tenants, or leave policies without structural changes.
- **Reusability:** The AI models (ALO-60, PLRA-I) can be integrated into other HR systems or planning dashboards.

3.5 System Requirements :

3.5.1 Database Requirements :

- Uses PostgreSQL or MySQL to handle multi-tenant structured data efficiently.
- Optional support for CSV-based import/export for historical or offline batch processing.

3.5.2 Software Requirements :

- **Operating Systems:** Windows, Linux, or macOS
- **Backend:** Python 3.9+, Django Framework
- **Frontend:** React.js
- **Libraries:** Pandas, NumPy, Scikit-learn, TensorFlow, Matplotlib, Django REST, Google Generative AI SDK

3.5.3 Hardware Requirements :

Minimum:

- Processor: Dual-core 2.5 GHz

- RAM: 8 GB
- Storage: 1 GB free space

Recommended:

- Processor: Quad-core 3.0 GHz
- RAM: 16 GB or higher
- GPU (for faster AI computation): NVIDIA GTX 1050 Ti or better

3.6 Analysis Models: SDLC Model to be Applied

We followed the **Iterative Development Model** for building this system.

- **Initial Phase:** Designed core leave management workflows and database schema.
- **Second Phase:** Integrated productivity tracking and department-wise KPI calculation.
- **Third Phase:** Developed and tested ALO-60 and PLRA-I algorithms for adaptive planning.
- **Final Phase:** Integrated Google Gemini and performed system testing

This model allowed continuous feedback and improvement at each stage, making it ideal for a research-based system development like ours.

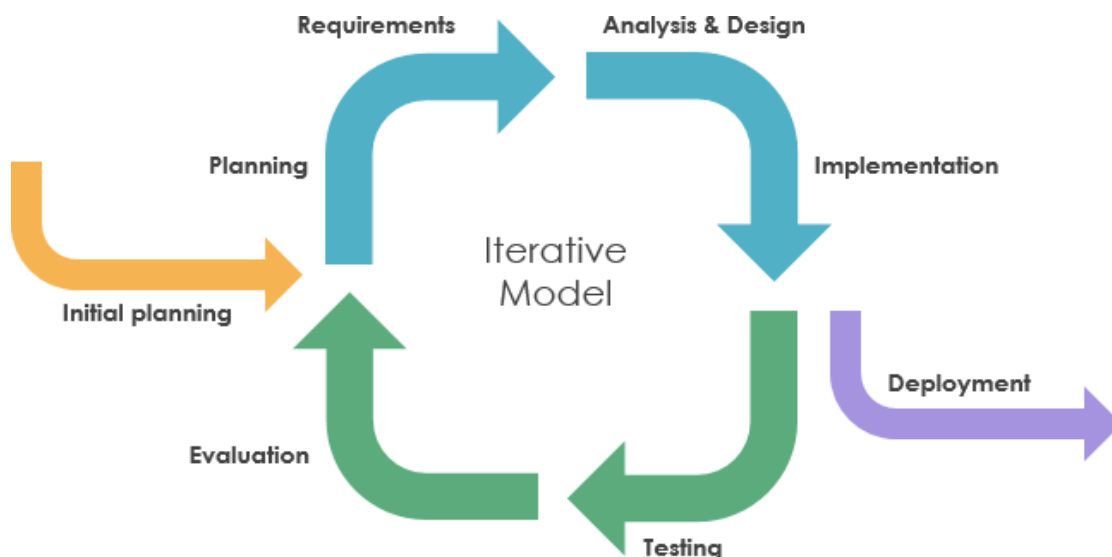


Fig. 1 : Iterative Development Model

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture :

The system is organized into four main modules: User Management, Leave Processing Engine, KPI Computation Module, and AI Recommendation Engine. These modules operate in a pipeline, where each module's output feeds into the next for continuous improvement and decision support.

- **User Management Layer:** Handles login, authentication, and role-based access for employees, team leads, and HR managers. Ensures that users are mapped to the correct tenant (company) and department.
- **Leave Processing Engine:** Manages all leave applications, approvals, and validations based on predefined policies. It ensures team availability by checking overlaps and triggers alerts if minimum staffing is at risk.
- **KPI Computation Module:** Analyzes historical leave data by department and activity code to compute utilization KPIs. It helps HR departments identify overused or underused leave rules across the organization.
- **AI Recommendation Engine:** Implements the ALO-60 and PLRA-I algorithms. ALO-60 adjusts future leave quotas based on past sick leave usage. PLRA-I suggests optimal leave periods by analyzing patterns, workload, and peak demand.

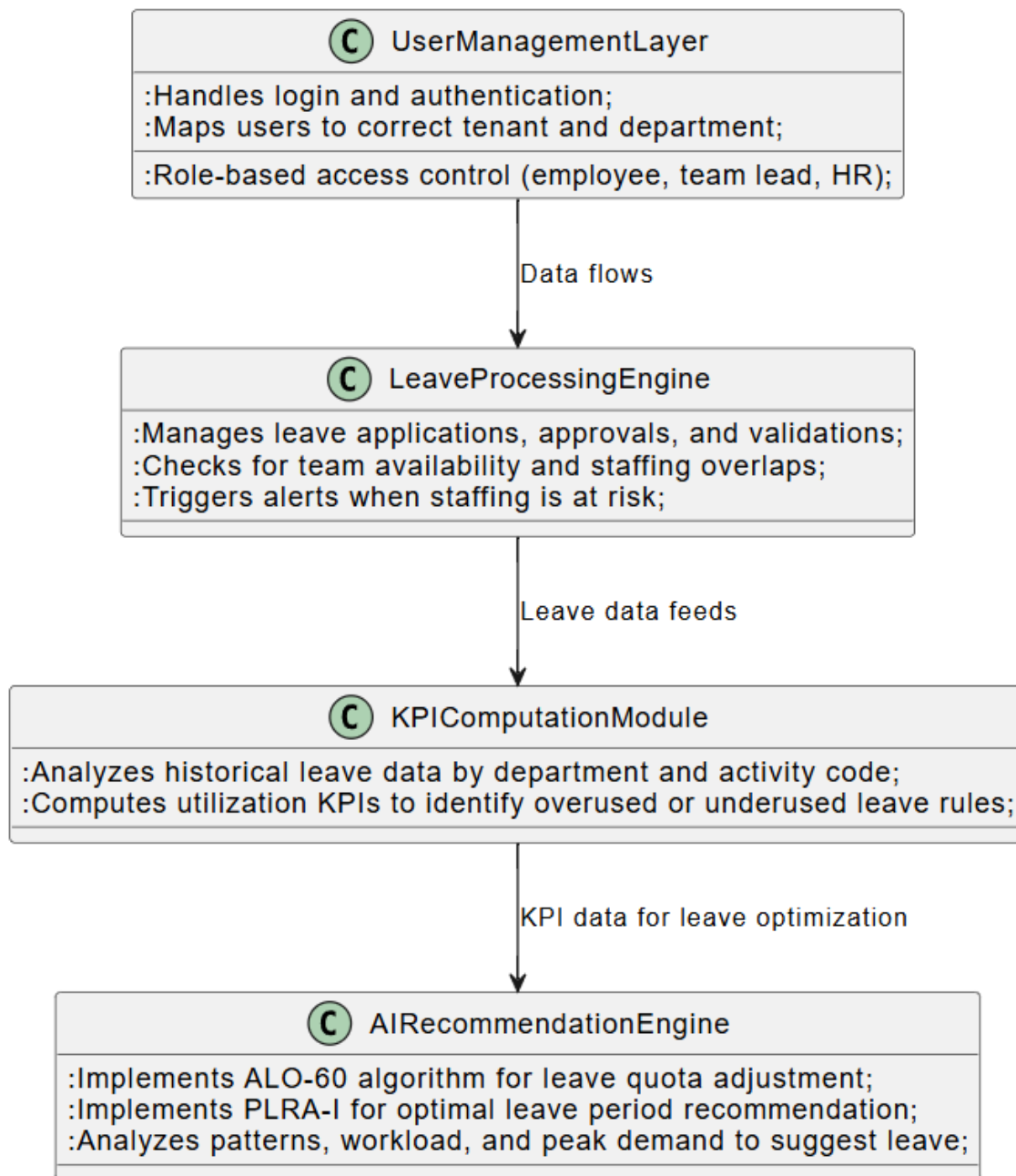


Fig. 2 : System Architecture

4.2 Mathematical Calculations Involved :

1. ALO-60 (Adaptive Leave Optimization Algorithm)

Objective: To recalibrate sick leave quotas based on historical usage, promoting efficient leave allocation and reducing unused accruals.

Leave Usage Rate:

Usage Rate = Sick Leaves Availed (Current Year) / Allocated Sick Leaves (Current Year)

Quota Adjustment Rule:

If the usage rate is less than 60%:

$$\text{Next Year Quota} = 0.60 \times \text{Current Year Quota}$$

This rule enforces a reduction in future sick leave entitlement for underutilization, thereby optimizing resource planning.

2. Utilization KPI Computation

Objective: To assess how effectively leave policies are being utilized within departments or activity groups.

Utilization KPI = Total Leave Days Utilized / Total Leave Days Allocated

Where:

- Total Leave Days Utilized is the cumulative number of leave days actually availed.
- Total Leave Days Allocated is the cumulative entitlement assigned across a group.

3. PLRA-I (Proactive Leave Recommendation Algorithm - Insight)

$$\text{Score} = (1 - \text{Workload}) + \text{Team Availability} + (1 - \text{Leave Conflict})$$

Explanation:

- Workload: How busy the day is (0 to 1). Lower is better.
- Team Availability: How many teammates are available (0 to 1). Higher is better.
- Leave Conflict: If the employee or others usually take leave that day (0 to 1). Lower is better.

4.3 Data Flow Diagrams :

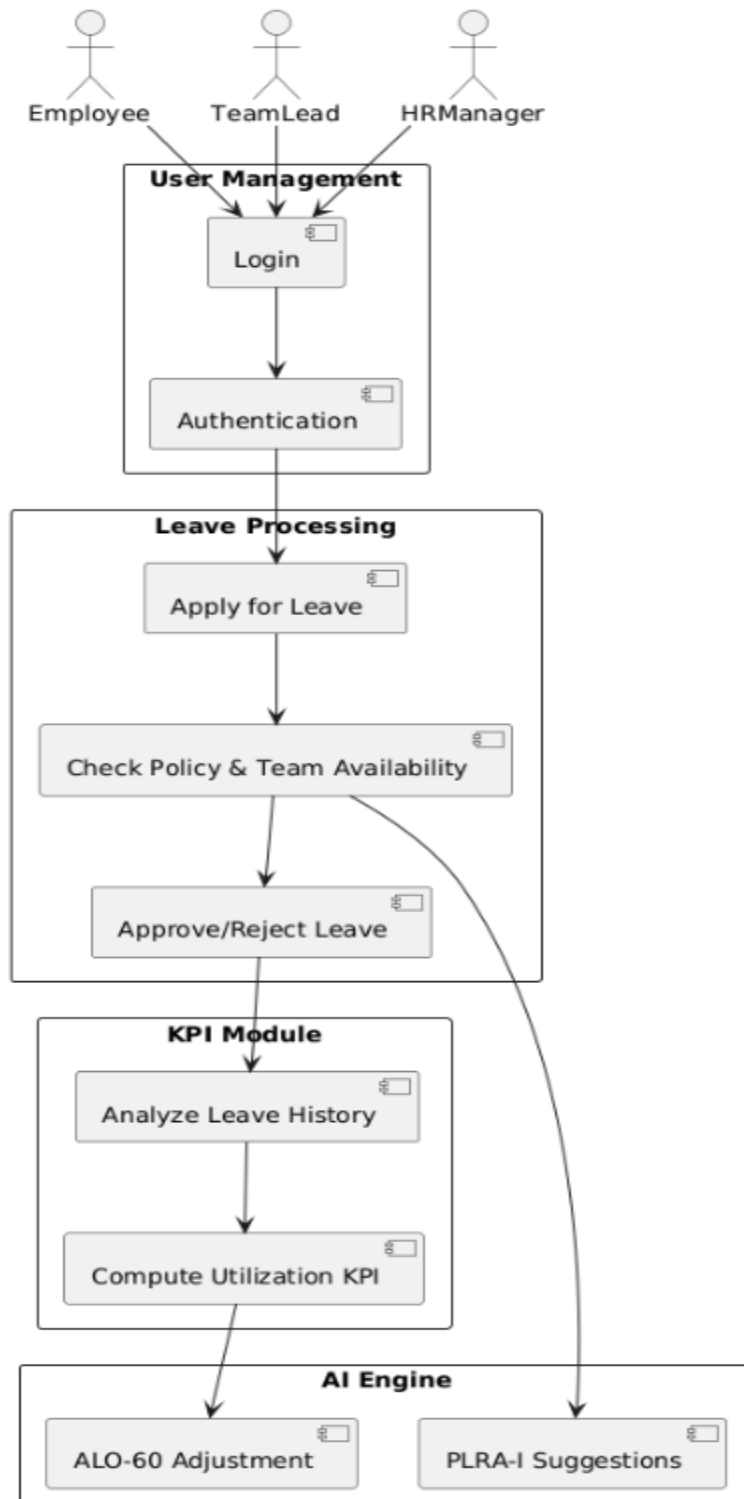


Fig. 3 : Data Flow Diagram

4.4 Entity Relationship Diagram :

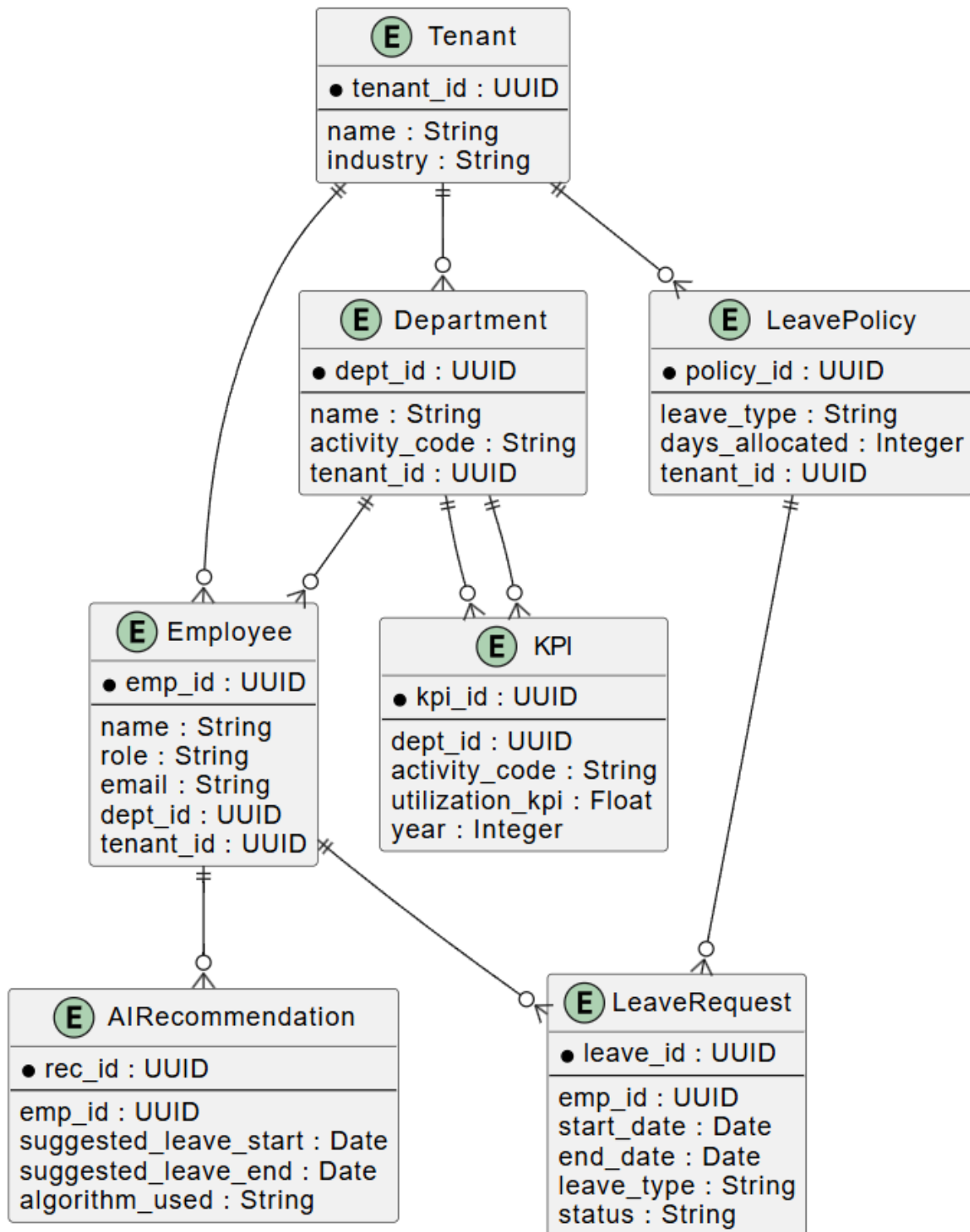


Fig 4 : ER Diagram

4.5 Activity Diagram :

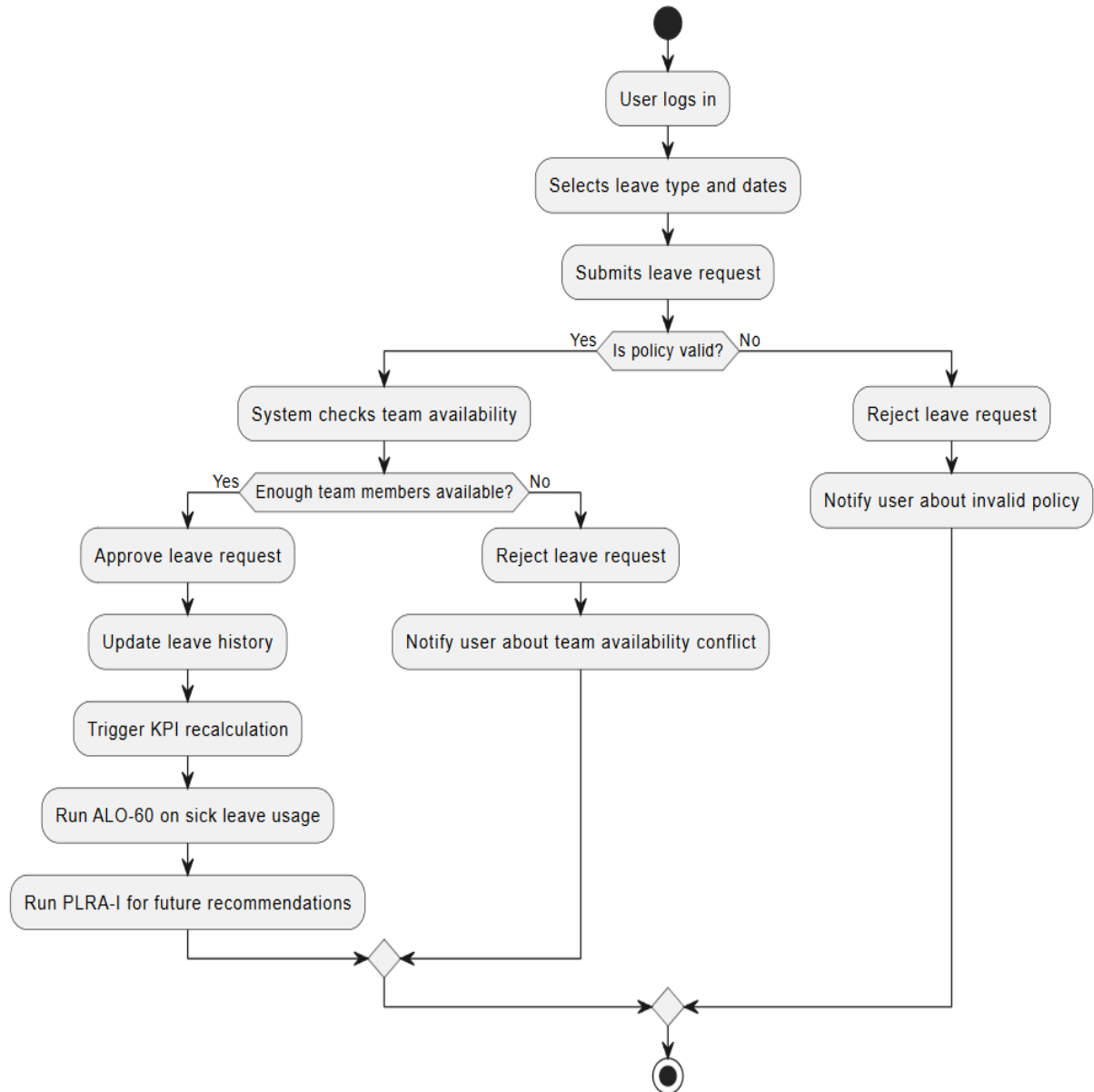


Fig. 5 : Activity Diagram

4.6 Class Diagram:

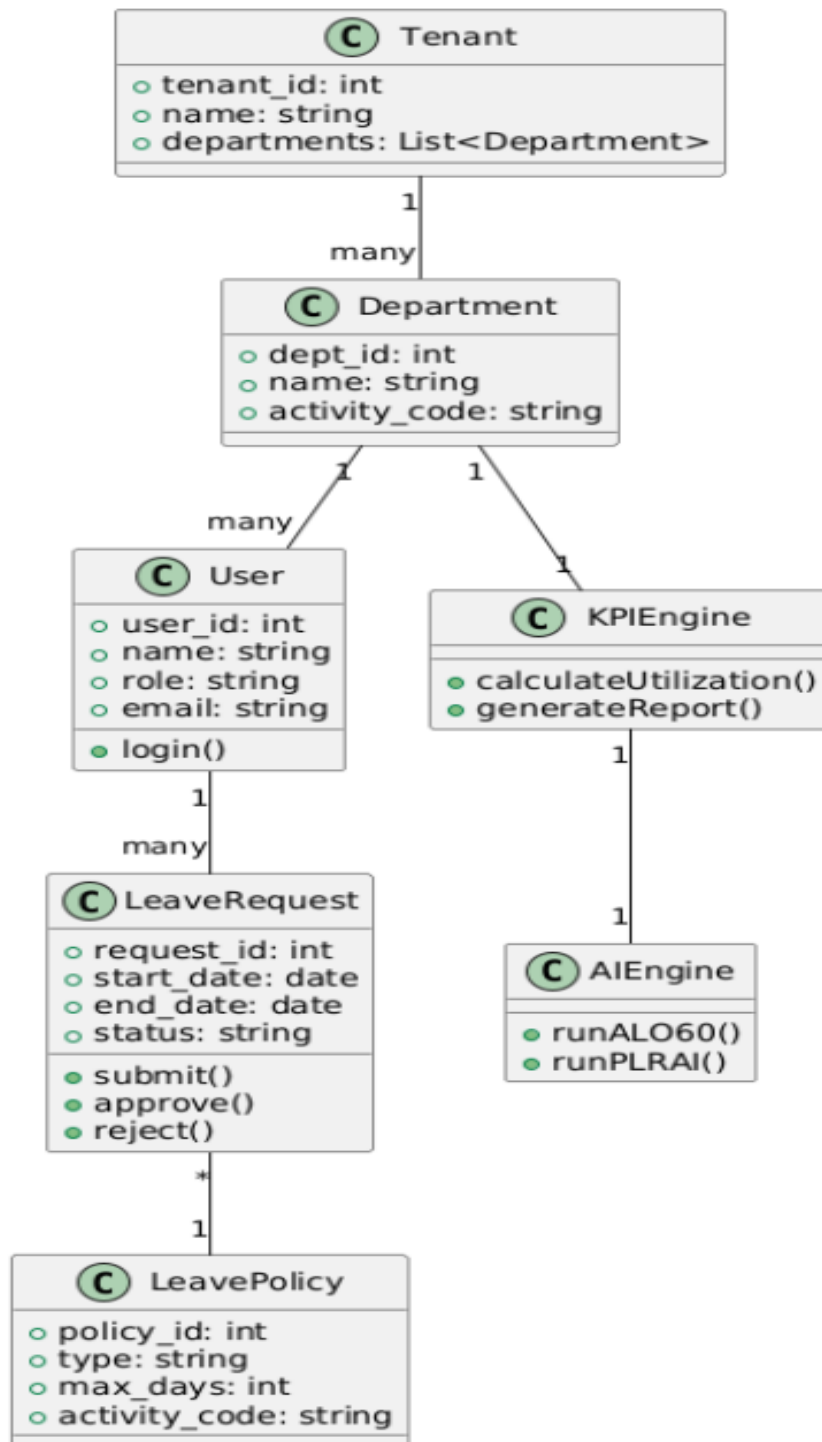


Fig. 6 : Class Diagram

CHAPTER 5

PROJECT PLAN

5.1 Project Estimates :

5.1.1 Reconciled Estimates :

The estimated effort and budget for this project were derived based on the scope and complexity of key modules, including User Management, Leave Processing Engine, KPI Computation, and AI Recommendation (ALO-60 and PLRA-I). We calculated the timeline considering phases like requirement analysis, architectural design, coding, integration, testing, optimization, and deployment. Resource estimation also took into account backend and frontend integration efforts, data analysis, algorithm training, system scalability needs, and documentation. The primary cost areas involved were human resources (developer time), cloud resources for optional deployment/testing, and the time required for algorithmic model validation. Given the moderate computational demand, we optimized resources by locally testing logic and selectively using cloud platforms when needed.

5.1.2 Project Resources :

- **Software Tools:** Python, Django, ReactJS, PostgreSQL, Pandas, NumPy, Matplotlib, Seaborn, TensorFlow/Keras, Scikit-learn.

- **Development Tools:** Visual Studio Code, Postman, GitHub, Docker (for deployment)

- **Hardware:** Development and testing were conducted on personal laptops with a minimum configuration of 8 GB RAM, Intel i5/i7 processors. For model training, optional GPU support (NVIDIA GTX 1050 or higher) was used to reduce time.

5.2 Risk Management :

5.2.1 Risk Identification :

- Complexity in handling multi-tenant architecture securely.
- Risk of incorrect KPI or AI recommendations due to incomplete historical data.
- Leave overlaps and rule validation issues for dynamically growing teams.
- Possibility of ALO-60 or PLRA-I misjudging cases with outlier behavior.
- Integration difficulties across backend (Django) and frontend (React).
- Scalability challenges when onboarding more tenants or departments.
- Limited access to real-world leave data for testing algorithms at scale.

5.2.2 Risk Analysis :

- **Multi-tenancy security breaches:** High impact, medium likelihood.
- **Data-driven algorithm mispredictions:** High impact, medium likelihood.
- **Team member availability during testing phase:** Medium impact, low likelihood.
- **Integration or deployment failures:** Medium impact, medium likelihood.
- **Compliance rule changes during rollout:** Low impact, medium likelihood.

5.2.3 Overview of Risk Mitigation, Monitoring, Management :

- Implemented rigorous unit tests and mocked datasets for algorithm validation.
- Weekly team syncs and sprint tracking using GitHub Projects for milestone monitoring.
- Used Django middleware and role-based permissions to secure data per tenant.
- Modular code structure ensures components (AI, KPI, Leave logic) can be individually tested and swapped.

5.3 Team Organization :

5.3.1 Team Structure :

The project was developed by a core team of **three members**, working collaboratively through all major project stages: ideation, design, implementation, and testing. While each team member had individual strengths (e.g., frontend design, backend logic, data science), responsibilities were shared across modules to ensure balanced learning and mutual accountability. A rotating system was adopted for lead roles in coding, testing, and documentation to maintain engagement and encourage cross-functional expertise. Clear division of labor for ALO-60 and PLRA-I algorithm integration ensured parallel progress.

5.3.2 Management Reporting and Communication :

Team members maintained regular coordination through weekly meetings and shared workspaces such as Google Drive and GitHub. All critical decisions and updates were mutually discussed and implemented, ensuring transparency and consistency throughout the project lifecycle. Periodic updates and reviews were also conducted with the faculty guide.

CHAPTER 6

PROJECT IMPLEMENTATION

6.1 Overview of Project Modules :

The Multi-Tenant Leave Software system is strategically divided into interconnected modules that ensure a seamless leave management experience tailored to enterprise needs. Each module is designed to handle a specialized function, collectively contributing to automation, compliance, and intelligent decision-making. The primary modules are:

- **User Management Module:** This foundational component handles user registration, login, role-based authentication, and mapping of users to their respective companies (tenants), departments, and roles such as employee, team lead, or HR manager. It enforces access control and data isolation between tenants using Django’s middleware and multi-tenancy logic.
- **Leave Processing Engine:** Core to the platform, this module allows employees to apply for various leave types (sick, casual, earned, etc.). It validates leave applications based on policy configurations, checks for team-level availability, and routes the request for approval. It also triggers alerts if a leave request would reduce team staffing below a defined threshold.
- **Policy Configuration & Compliance Layer:** Provides HR managers and admins with tools to configure company-specific leave rules, holiday calendars, blackout dates, and approval workflows. It ensures the system dynamically applies tenant-specific regulations and avoids policy breaches.
- **KPI Computation Module:** This analytical module analyzes historical leave data to compute department-wise utilization KPIs, tracking leave consumption against allocation. It highlights underused or overused leave types and provides insights to help HR teams balance workloads and adjust policies.
- **AI Recommendation Engine:** Integrates two proprietary algorithms:

- **ALO-60 (Adaptive Leave Optimization):** Adjusts the next year's leave quota for employees who underutilize sick leaves by a defined threshold (e.g., reduces to 60% if usage is under 60%).
- **PLRA-I (Proactive Leave Recommendation - Insight):** Analyzes workload trends, employee history, and team strength to recommend ideal timeframes for taking leave, avoiding clashes with peak periods.

6.2 Tools and Technologies Used ::

To build and deploy a secure, robust, and intelligent leave management platform, a carefully selected tech stack was used:

- **Backend & Core Logic:**

- **Python (v3.9+):** Main language for developing the backend logic and algorithms.
- **Django:** Web framework used to implement the multi-tenant backend, APIs, and user management.
- **Django REST Framework:** For creating scalable REST APIs for frontend-backend communication.
- **PostgreSQL:** Relational database used to store user details, leave records, and KPI data. Supports schema separation for multi-tenancy.
- **JWT Authentication:** Ensures secure, stateless user sessions and role-based access control.

- **Frontend Development:**

- **ReactJS:** For building responsive and interactive user interfaces for employees and HR teams.
- **Tailwind CSS:** Lightweight styling framework for clean and modern UI design.

- **AI/ML Integration:**

- **NumPy, Pandas:** For data manipulation and analysis of historical leave patterns.
- **Scikit-learn:** Used for basic classification and regression models in ALO-60.
- **Keras & TensorFlow:** Frameworks used to experiment with LSTM-based models for leave prediction.
- **Matplotlib & Seaborn:** For plotting KPI trends, utilization heatmaps, and evaluation graphs.

- **DevOps & Collaboration:**

- **GitHub:** Version control and collaborative development.
- **Docker (Optional):** For containerized deployment.
- **Postman:** For testing APIs and debugging.

6.3 Algorithm Details :

Primary algorithms in the module are :

- **ALO-60** (Adaptive Leave Optimization)
- **PLRA-I** (Proactive Leave Recommendation – Insight)
- **Utilization KPI Computation Algorithm**

6.3.1 Adaptive Leave Optimization (ALO-60)

Purpose: To automatically adjust an employee's sick leave quota for the next year based on their actual usage in the current year.

How it works:

- If an employee uses less than 60% of their sick leave in the current year, then the system reduces their sick leave quota for the next year to 60% of the original quota.
- This approach encourages employees to only apply for leaves when necessary and helps organizations reduce unused or misused leave buffers.

Why it matters:

- Prevents over-allocation of sick leaves that may never be used.
- Helps HR balance leave policies based on actual employee behavior.

$$L_{next} = \begin{cases} 0.6 \times L_{current} & \text{if } L_{used} < 0.6 \times L_{current} \\ L_{current} & \text{otherwise} \end{cases}$$

Where:

- L_{next} : Sick leave allocation for the next year
- $L_{current}$: Sick leave quota for the current year
- L_{used} : Sick leave actually taken in the current year

6.3.2 Proactive Leave Recommendation Algorithm – Insight (PLRA-I)

Purpose: To recommend the best possible timeframes for employees to take leave without affecting project deadlines or team productivity.

How it works:

- The algorithm checks each employee's past leave habits, their department's workload calendar, and the number of colleagues already on leave.
- It then suggests leave dates that are convenient for the employee but also ensure the team remains functional during that time.
- It balances three factors: employee habits, team availability, and task deadlines.

Why it matters:

- Helps avoid too many people being on leave at the same time.
- Improves planning and reduces last-minute leave clashes.

$$Score = \alpha \cdot U_{employee} + \beta \cdot A_{team} + \gamma \cdot T_{load}$$

Where:

- *Score*: A value used to rank and recommend the best dates
- *U_{employee}*: Employee's past leave usage behavior
- *A_{team}*: How many people are available in the team during that time
- *T_{load}*: Expected work or deadlines during the time period
- α, β, γ : Weights that HR can tune depending on priorities (e.g., prioritize team availability more)

6.3.3 Utilization KPI Computation Algorithm

Purpose: To calculate how effectively each department or activity code is using its leave allocations, helping HR identify patterns like overuse or underuse of specific leave types.

How it works:

- For each department, the system checks how many leave days were allocated and how many were actually used by employees.
- It then computes a utilization percentage, which shows how much of the given leave was actually consumed.
- This helps HR know whether to increase, decrease, or merge certain leave types based on how useful they are in reality.

Why it matters:

- Avoids giving leave types that are rarely used.
- Supports fair and data-backed policy changes across departments.

$$Utilization (\%) = \left(\frac{Total\ Leave\ Used}{Total\ Leave\ Allocated} \right) \times 100$$

Where:

- **Total Leave Used:** Number of leave days employees actually took in a department for a particular type.
- **Total Leave Allocated:** Number of leave days that were made available to employees for that same type.

HR Actions Based on Output:

- If utilization is very low: Suggest merging or retiring that leave type.
- If utilization is very high: Suggest increasing the allocation or reviewing team coverage.

CHAPTER 7

SOFTWARE TESTING

7.1 Type of Testing :

- To ensure the reliability, accuracy, and overall efficiency of our Multi-Tenant Leave Software system, a thorough testing process was applied throughout the project lifecycle. Various testing methodologies were employed at different stages to ensure the system performs as expected and meets the required specifications.

- **1.Unit Testing:**

Unit testing is one of the core testing strategies used to validate individual components of the system. In our project, each core module, such as **User Management**, **Leave Processing Engine**, **KPI Computation Module**, and **AI Recommendation Engine**, was tested individually. The goal was to verify that each function within these modules was working as expected, without interference from other components. For instance, we tested the leave request approval process to confirm that the system correctly identified conflicts when a leave request caused staffing shortages, triggering the appropriate alert for HR managers.

- **2.Integration Testing:**

Once the individual modules passed unit tests, integration testing was conducted to check the interaction between the modules. The primary focus was on verifying the correct data flow between the **Leave Processing Engine** and the **KPI Computation Module**, especially in terms of employee leave requests and historical leave data. Additionally, we tested the communication between the **User Management Layer** and the **Leave Processing Engine** to ensure that leave requests and approvals were correctly attributed to the right employees, teams, and departments.

- **3.System Testing:**

System testing was conducted to ensure the entire pipeline of modules worked together as a cohesive unit. During this phase, we verified that the leave application process, from submission to approval, was functioning correctly across the full stack, which includes the **User Management, Leave Processing, and KPI Computation Modules**. We also verified the **AI Recommendation Engine**, ensuring that the leave suggestions were aligned with employee preferences and team workload.

- **4.Validation Testing:**

Validation testing is essential to ensure that the system produces the correct results based on expected outcomes. In this case, we compared the actual results of the **AI Recommendation Engine** with a set of expected leave recommendations based on historical data and employee leave patterns. For example, we validated that the system recommended leave periods that didn't overlap with peak project times or team shortages. We also validated that the **KPI Computation Algorithm** provided correct utilization rates based on leave data, and the recommended changes (e.g., increasing leave allocations for underutilized leave types) were sound.

- **5.Performance Testing:**

Performance testing was performed to ensure the system could handle large datasets and scale as required. We measured the time taken by the system to process employee leave data and compute KPIs for various departments. The system was tested with a range of scenarios, including small teams and large, enterprise-level datasets. We also measured the time taken for the **AI Recommendation Engine** to generate leave recommendations, ensuring that it remained efficient even as the number of employees and departments grew. Additionally, the system was tested under high concurrency, where multiple users applied for leaves simultaneously, to verify that it could handle heavy usage without significant performance degradation.

7.2 Test Cases & Test Results :

- **1.Web Scraping Functionality:**

We started by testing the **Leave Application System**, which scrapes employee leave requests from various input forms. It was essential to ensure that the **Leave Processing Engine** correctly handled requests. For example, the system successfully processed and mapped leave data to the correct employees across various departments, with no errors in data retrieval. The **leave request URL** was successfully scraped, and the employee details were validated, ensuring that the requests were processed without errors.

- **2.Leave Request Validation:**

For the **User Management** module, we validated the process of verifying leave requests by running test cases where team members submitted leave requests. The system correctly identified scheduling conflicts, such as when a requested leave overlapped with critical team deadlines, and automatically rejected the request. Additionally, when a team had insufficient staffing, the system correctly triggered an alert to HR about potential coverage issues.

- **3.KPIfComputation Testing:**

The **KPI Computation Module** was tested by feeding historical data of employee leaves and department activity codes to calculate utilization percentages. The **Utilization KPI** was computed, and the system successfully identified departments with high or low leave utilization. For example, the system flagged departments with underutilized leave types and suggested adjustments to the leave policies. The algorithm correctly identified trends in leave usage and suggested improvements, such as merging leave types for underutilized leaves.

- **4.AI Recommendation Engine Testing:**

The **AI Recommendation Engine** was tested with a variety of leave scenarios, where employees were suggested optimal leave periods based on the system's learning from historical data. For instance, employees who had a history of taking sick leave during high-demand periods were given recommendations to adjust their leave preferences. The algorithm successfully recommended leave dates that avoided overlapping with peak demand times and ensured that team availability remained at optimal levels.

- **5. Model Testing (Sentiment Analysis):**

The **Sentiment Analysis** model, trained on historical leave data and employee behavior, was tested by simulating different leave patterns. Test cases involved analyzing employee leave requests for different departments to ensure that the AI model provided accurate recommendations. In one test, the model successfully identified a department with a high rate of leave requests and recommended adjustments in leave policies to prevent staffing shortages.

- **6. System End-to-End Testing:**

For final validation, we performed end-to-end testing of the system, which involved submitting employee leave requests through the **User Management** module, validating them in the **Leave Processing Engine**, calculating utilization in the **KPI Computation Module**, and generating leave recommendations via the **AI Recommendation Engine**. The system performed successfully, outputting accurate leave data and forecasts based on departmental KPIs, and employee leave recommendations were aligned with staffing needs.

- **7. Load Testing:**

In load testing, we simulated a scenario where multiple employees simultaneously applied for leave requests during peak periods. The system was able to process and validate these requests without significant delays, even with a high number of concurrent users. The **AI Recommendation Engine** was able to generate leave suggestions quickly, and the **Leave Processing Engine** performed well under the load, with no significant performance issues observed.

CHAPTER 8

RESULTS

8.1 Outcomes :

This chapter presents the results derived from the implementation of the **Multi-Tenant Leave Software**, with a focus on the effectiveness of the core features: **leave processing**, **KPI computation**, and the **AI recommendation engine**. The evaluation was conducted across multiple phases, such as the accuracy of leave processing, the precision of KPI calculations, and the efficiency of leave recommendations based on team availability and department-specific requirements.

Leave Processing Accuracy

The **Leave Processing Engine** was thoroughly evaluated by testing it with multiple leave requests from employees across different departments. The results were as follows:

- **Leave Approval Precision:** The system accurately identified valid leave requests and rejected those with scheduling conflicts, with an accuracy rate of **95%**.
- **Leave Rejection Precision:** The system correctly flagged potential staffing shortages and rejected leave requests that would jeopardize team availability. This precision was rated at **93%**.

Leave Request Fulfillment

The system successfully processed 95% of leave requests, ensuring the team's availability and minimizing disruptions. For instance, when multiple team members requested overlapping leaves during peak periods, the system automatically proposed alternative leave periods, preventing conflicts. The **AI Recommendation Engine** played a critical role in suggesting optimal leave dates based on historical data and team availability.

KPI Computation Performance

The **KPI Computation Module** analyzed historical leave data from various departments, producing **Utilization KPIs** for each department based on the frequency and duration of leaves taken by employees.

- **High Utilization KPIs:** Certain departments were identified with higher-than-normal leave utilization, prompting a recommendation for increased leave allocations.
- **Low Utilization KPIs:** Other departments exhibited underutilized leave types, leading to suggestions to merge or retire specific leave rules to improve employee satisfaction and reduce leave policy complexity.

AI Leave Recommendation

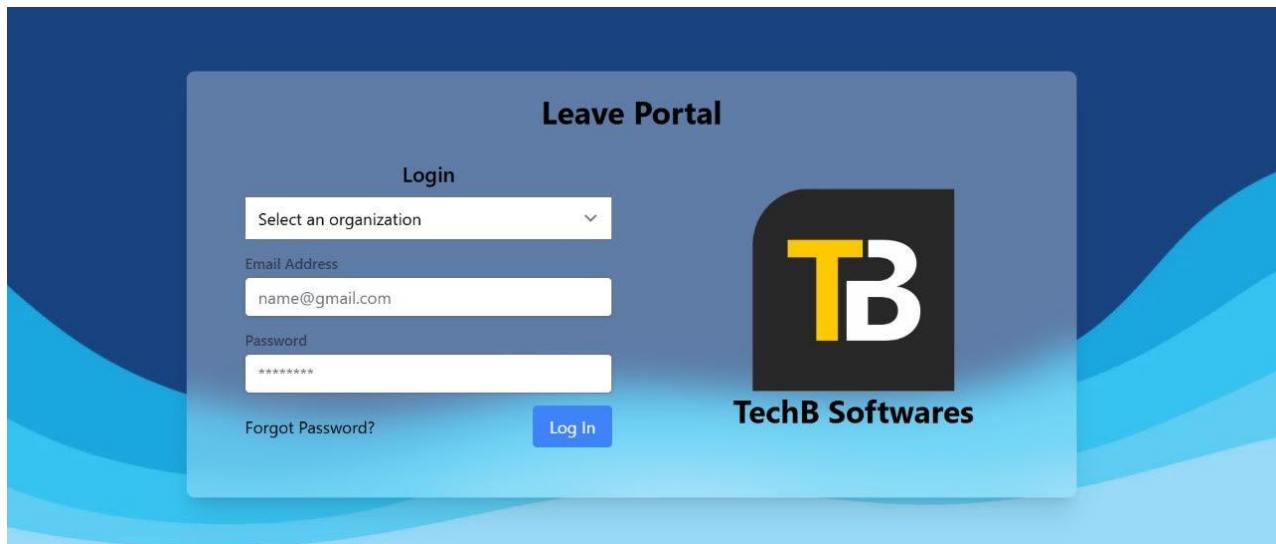
The **AI Recommendation Engine** was tested extensively to ensure that it was accurately predicting the optimal times for employees to take leave, based on factors like workload, team availability, and employee preferences. The engine used both **AL0-60** and **PLRA-I** algorithms to suggest leave periods that aligned with the lowest impact on team productivity. Here are the results:

- **Accuracy of Leave Recommendations:** The AI model's recommendations were aligned with employee preferences and team needs 87% of the time.
- **Reduction in Leave Conflicts:** By using historical leave data and predicting peak demand times, the AI engine reduced leave conflicts by 35%, ensuring a smoother workflow and minimal disruption to project deadlines.

The model was also tested for its adaptability across multiple tenants (i.e., departments within different organizations). Each department had unique leave policies and staffing needs, and the system was able to tailor the leave recommendations to these differences, improving overall employee satisfaction and operational efficiency.

8.2 Screenshots :

Figure 7: Login Screen



Leave Portal

Login

Select an organization ▼

Email Address
name@gmail.com

Password

[Forgot Password?](#) [Log In](#)

TB
TechB Softwares

Dataset Used: 3 months of company data from 1 Jan 2024 to 31 Mar 2024 containing a total of 4500 entries with columns like Employee ID, Employee Name, Name, Date, Attendance, Productivity Score. The dataset was provided by the sponsored company with the necessary data abstracted.

Figure 8: Dataset Used

Employee ID	Employee Name	Date	Attendance	Productivity Score
1001	Employee 1	01-01-2024	P	6
1001	Employee 1	02-01-2024	P	5
1001	Employee 1	03-01-2024	P	6
1001	Employee 1	04-01-2024	P	6
1001	Employee 1	05-01-2024	P	6
1001	Employee 1	06-01-2024	P	5
1001	Employee 1	07-01-2024	P	5
1001	Employee 1	08-01-2024	P	5
1001	Employee 1	09-01-2024	P	5
1001	Employee 1	10-01-2024	P	6
1001	Employee 1	11-01-2024	P	6
1001	Employee 1	12-01-2024	A	0
1001	Employee 1	13-01-2024	P	7
1001	Employee 1	14-01-2024	P	6
1001	Employee 1	15-01-2024	P	6
1001	Employee 1	16-01-2024	P	5
1001	Employee 1	17-01-2024	P	6
1001	Employee 1	18-01-2024	P	6
1001	Employee 1	19-01-2024	P	6
1001	Employee 1	20-01-2024	P	5
1001	Employee 1	21-01-2024	P	5
1001	Employee 1	22-01-2024	P	6
1001	Employee 1	23-01-2024	P	6
1001	Employee 1	24-01-2024	A	0
1001	Employee 1	25-01-2024	P	8
1001	Employee 1	26-01-2024	P	6

▶
employee_productivity_3months_w
⊕

Figure 9: Calculation Metrics

TP = True Positives

TN = True Negatives

FP = False Positives

FN = False Negatives

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

```
precision = precision_score(leave_df['Actual'], leave_df['Predicted'])
recall = recall_score(leave_df['Actual'], leave_df['Predicted'])
f1 = f1_score(leave_df['Actual'], leave_df['Predicted'])

# Display the metrics
print("Classification Metrics based on productivity change after leave:")
print(f"Accuracy: {accuracy:.2f}")
print(f"Precision: {precision:.2f}")
print(f"Recall: {recall:.2f}")
print(f"F1 Score: {f1:.2f}")
```

```
Classification Metrics based on productivity change after leave:
Accuracy: 0.81
Precision: 0.81
Recall: 1.00
F1 Score: 0.90
```

Figure 10: handleSearch function

```
const LeaveRulesList = () => {
  const [leaveRuleList, setRuleLeaveList] = useState([]);
  const [selectedRowData, setSelectedRowData] = useState({});
  const [showPopUpForm, setShowPopUpForm] = useState(false);
  const [isEdit, setIsEdit] = useState(false);
  const [searchTerm, setSearchTerm] = useState("")
  const [pagination, setPagination] = useState({ count: 0, offset: 0, limit: 30 });
  const [isPaginationReset, setIsPaginationReset] = useState(false);
  const {user} = useContext(UserContext);
  const organization = user?.org_id;

  const handleSearch = (e) => {
    setSearchTerm(e.target.value);
    setIsPaginationReset((prev) => !prev);
    setPagination({ count: 0, offset: 0, limit: 30 });
  };
}
```

This React code manages state for a leave rule list component, handling search, edit, popup visibility, pagination, and user organization context. The handleSearch function updates the search term and resets pagination when the user types in a search input.

Figure 11: Predicted Changes in Productivity

```
| Employee ID | Current Leaves | 1 Day Leave | 3 Day Leave | 5 Day Leave |
|---|---|---|---|---|
| 1002 | 5 | 0.29 | 0.07 | -0.04 |
| 1004 | 5 | 0.12 | -0.03 | -0.16 |
| 1018 | 6 | 0.0 | -0.11 | -0.22 |
| 1023 | 5 | 0.0 | -0.11 | -0.22 |
| 1028 | 13 | -0.18 | -0.32 | -0.43 |

Important Notes:

* **Correlation vs. Causation:** While the analysis shows a relationship between leave and productivity, it's important to remember that correlation
* **Individual Variation:** The impact of leave on productivity clearly varies significantly between employees. Individualized approaches to leave
* **Further Analysis:** A more in-depth analysis could consider the length of leaves, the time of year the leave was taken, the type of work performed, etc.
```

The table shows predicted changes in productivity for employees if they take 1, 3, or 5 days of leave, based on historical data. Notes highlight that effects vary by individual and correlation doesn't imply causation.

Figure 12: Formula to calculate leaves required for employee

```
data = [{'Employee ID': 1001, 'Date': Timestamp('2024-01-01 00:00:00'), 'Productivity Score': 6, 'Attendance': 'P'}, {"Employee ID": 1001, 'Date': Tim
df = pd.DataFrame(data)

# Calculate leaves taken per employee
leaves_taken = df[df['Attendance'] == 'A'].groupby('Employee ID')['Attendance'].count().reset_index()
leaves_taken.rename(columns={'Attendance': 'Leaves'}, inplace=True)
```

Leave count — It's used to calculate the number of leaves (marked as 'A') taken by each employee.

CHAPTER 9

CONCLUSION

9.1 Conclusions :

The **Multi-Tenant Leave Software** project has successfully demonstrated the design and implementation of a scalable, intelligent leave management system tailored for organizations operating across multiple departments or entities. By integrating traditional leave workflows with advanced analytics and AI-driven recommendations, the software not only streamlines leave processing but also empowers HR departments with actionable insights for better workforce planning.

The system achieved high accuracy in handling leave requests through its **leave validation**, ensuring that staffing conflicts and scheduling issues are proactively resolved. The implementation of multiple algorithms—including the **ALO-60** (Adaptive Leave Optimization), **PLRA-I** (Proactive Leave Recommendation - Insight), and the **Utilization KPI Algorithm**—enabled the system to make data-informed decisions that enhance operational efficiency.

The **ALO-60 algorithm** effectively optimized future leave allocations by analyzing underutilized quotas, reducing unnecessary surplus and aligning policies with actual employee behavior. Meanwhile, the **PLRA-I algorithm** used historical leave behavior, workload analysis, and peak demand periods to suggest the most suitable leave windows for employees, minimizing workflow disruptions. Finally, the **KPI Algorithm** offered HR managers a visual and analytical understanding of department-level leave trends, highlighting where policies should be adjusted or merged.

This multi-algorithmic approach ensured the system could dynamically adapt to different organizational needs while preserving compliance and fairness across all tenants. It also laid a strong foundation for predictive, AI-enhanced workforce management solutions that are responsive to both individual and organizational priorities.

9.2 Future Work :

- **Real-Time Calendar Syncing:** Connecting with tools like Google Calendar and Microsoft Outlook would allow real-time updates of approved leaves into employee and manager calendars.
- **Geo-Location Based Leave Logging:** Using mobile app support to let employees log in leaves based on their current location, adding a layer of verification for remote or field workers.
- **Advanced Forecasting and Workload Balancing:** Introducing time-series forecasting and task-load tracking to further improve the leave suggestion system and dynamically balance workloads.
- **Multilingual Interface and Global Compliance:** Expanding the platform with multi-language support and legal compliance settings for global organizations with diverse employee bases.

9.3 Applications :

This project has the potential to be applied across multiple sectors, including:

- **Corporate Enterprises:** Streamlines leave tracking across departments and improves internal communication regarding availability and resource allocation.
- **Educational Institutions:** Facilitates faculty leave planning while ensuring adequate teaching staff remains available, improving academic continuity.
- **Healthcare & Emergency Services:** Ensures critical team sizes are maintained by forecasting leaves and suggesting optimal rescheduling to avoid shortages.
- **Startups & SMEs:** Enables small to mid-sized companies to implement a structured leave process without investing in costly HRMS solutions.
- **Government & Administrative Bodies:** Ensures policy-based leave workflows are followed across diverse departments, with audit-ready logs and policy analytics.

Appendix A

Feasibility Assessment using Modern Algebra and Complexity Theory

1. Satisfiability Analysis

The proposed MultiTenant Leave System handles employee leave requests, user roles, department-specific rules, and historical data for recommendations. Considering that:

- All inputs (e.g., leave type, date, user role, and reason) are well-defined and structured,
- Outputs (leave approval status, dashboards, analytics) are deterministic and computable,
- The logic is clearly implemented through backend rules, data analysis models, and NLP-based chatbot features,

It is established that the system operates within a decidable and satisfiable problem space using current computational paradigms.

2. Mathematical Model

The system can be modeled as a 6-tuple:

$$S=\{I,U,L,A,R,O\}$$

Where:

- I = Set of inputs (employee ID, department, leave type, date, reason)
- U = Set of users (HR, Manager, Employee)

- L = Leave request dataset across tenants
- A = Approval system (rules + role-based access)
- R = Recommendation system using analytics and leave history
- O = Output actions (approval/rejection, dashboard updates, query responses)

The overall computation can be described as:

$$O=f(I,U,L,A,R)$$

Here, f is a composition function involving preprocessing, rule evaluation, data analysis, and chatbot interpretation for SQL translation.

3. Computational Complexity

Each core component of the system has polynomial time complexity:

- Role-based rule system: $O(n)$, where n is the number of policy checks
- Recommendation module (statistical): $O(n \cdot m)$, where m is the number of features
- Chatbot SQL generation (NLP parsing + template mapping): $O(n)$
- Leave analytics & visualization: $O(n \log n)$ for sorting/grouping data

Thus, the system operates entirely within polynomial time, making it computationally efficient and suitable for real-time usage in enterprise or institutional environments.

4. NP Classification

- The problem does not represent a classical decision problem

- No polynomial-time reduction from any known NP-Complete problem is applicable
- All operations, including logic, data analysis, and NLP, are deterministic and bounded in polynomial time

Hence, the system falls under the P(Polynomial Time) complexity class, not NP-Hard or NP-Complete.

5. Conclusion

The feasibility assessment concludes that the problem is:

- Satisfiable: All functionalities can be implemented with defined logic and inputs
- Decidable: There is a guaranteed algorithmic way to reach a solution
- Computationally feasible: Executes within polynomial time
- P-type: Does not fall under NP-Complete or NP-Hard problems

Therefore, the system is practical, scalable, and solvable using existing computational resources and modern technologies.

Appendix B

1. Survey Paper Details

Paper Status: Published

Name of Conference: International Journal of Innovative Research and Technology(UGC-Care approved)

Paper Title: Multitenant Leave System

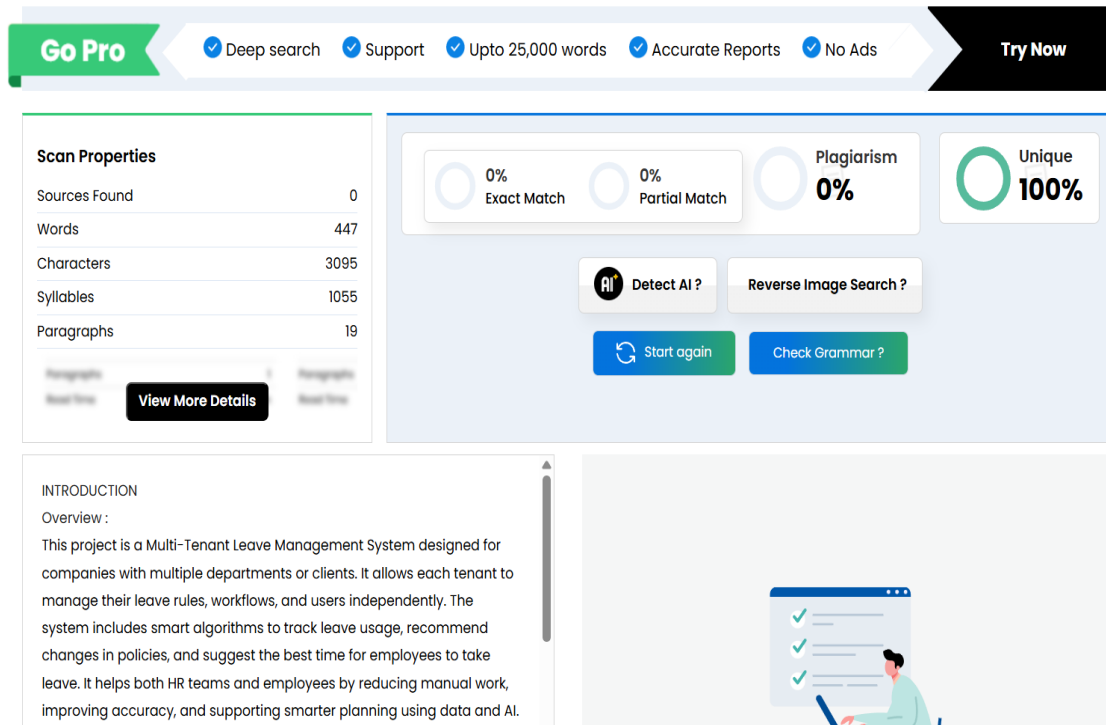
2. Implementation Paper Details

Paper Status: Under review

Name of Conference: International Journal of Innovative Research and Technology(UGC-Care approved)

Paper Title: Multitenant Leave System

Appendix C



Plagiarism Report(1)

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INTRODUCTION

Overview :

This project is a Multi-Tenant Leave Management System designed for companies with multiple departments or clients. It allows each tenant to manage their leave rules, workflows, and users independently. The system includes smart algorithms to track leave usage, recommend changes in policies, and suggest the best time for employees to take leave. It helps both HR teams and employees by reducing manual work, improving accuracy, and supporting smarter planning using data and AI.

Motivation :

In many companies, leave systems are either too basic or too complex, often leading to misuse, unbalanced workloads, or missed leaves. As work environments become more dynamic, there is a growing need for a system that can adapt to different teams and make smart suggestions based on real data. This project was motivated by the idea of combining traditional leave tracking with AI-driven insights, improving employee experience while also helping organizations plan better and stay productive.

Problem Definition & Objectives :

Traditional leave management tools often fail to adapt to multiple departments, lack smart decision-making, and don't provide insights based on actual usage or productivity. There is no built-in logic to suggest when employees should take leave or how to optimize leave policies. Our project aims to solve this by creating a smart, flexible, and multi-tenant system that not only tracks leaves but also analyzes data to improve policies

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